CSCI-570 Homework 4

Due Date: Tuesday, October 14th, 11:59pm

Problem 1.
Given a set of m positive integers $A = a_1, a_2, ..., a_m$, and a number n, provide a dynamic programming algorithm to find out whether it is possible to find a subset of A such that the sum of elements in this subset is n. Is the algorithm polynomial?

Example: $A = 1, 4, 12, 20, 9$ and $n = 14$, we can find a subset $1, 4, 9$ such that $1 + 4 + 9 = 14$.

[10 points]

Problem 2.
Imagine a robot sitting on the upper-left corner of an $m \times n$ grid. The robot can only move in two directions at each step: right or down. Write a program to compute the number of possible paths of the robot to the lower-right corner. What is the complexity of your program? (Assume that integer multiplication can be done in constant time!)

[6 points]

Problem 3.
Design an effective algorithm to compute what is the maximum number of characters in a given string that form a palindrome. (A palindrome is a word (a sequence of symbols) that can be read the same way in either direction, for example "*abaccaba*" is a palindrome.)

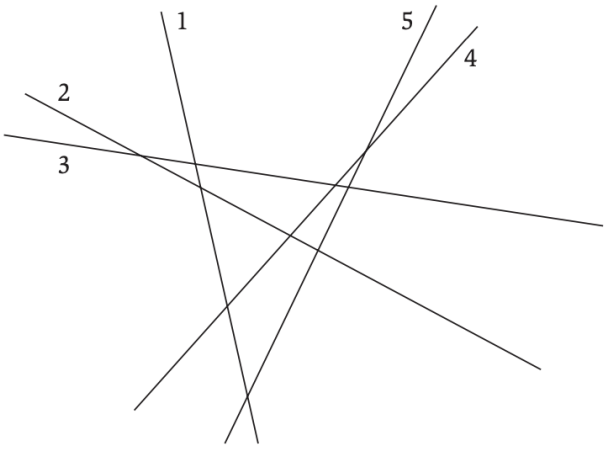
Example: in the word $bbacfabcbf$ the longest (non-contiguous) subsequences that form a palindrome are $bbafabb$ and $bbacabb$, both with length 7.

[12 points]

Problem 4.
Hidden surface removal is a very common problem in computer graphics. For example, in the movie *Toy Story*, when Woody is standing in front of Buzz, you should be able to see Woody, but not Buzz; when Buzz is standing in front of Woody..., you get the idea.

The magic of hidden surface removal is that you can often compute things faster than your intuition suggests. Here's a clean geometric example to to illustrate a basic speed-up that can be achieved. You are given n non-vertical lines in the plane, labeled $L_1, ..., L_n$, with the ith line specified by the equation $y = a_i \cdot x + b_i$. We will make the assumption that no three of the lines all meet at a single point. We say line L_i is *uppermost* at a given x-coordinate x_0 if its y-coordinate at x_0 is greater than the y-coordinates of all the other lines at x_0: $a_i \cdot x_0 + b_i > a_j \cdot x_0 + b_j$ for all $j \neq i$. We say line L_i is *visible* if there is some x-coordinate at which it is uppermost – intuitively, some portion of it can be seen if you look down from "$y = \infty$."

Give a divide and conquer algorithm that takes n lines as input and in $O(n \cdot log\ n)$ time returns all of the ones that are visible. For example, in the picture below all the lines except for 2are visible.

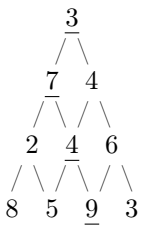


[Hints: order the lines according to their slopes.
Notice that the first and last line in this order will always be visible.
Look at a pair of lines, what is the relation between their visibility and slope?]

[12 points]

Problem 5.
Consider the triangle below constructed such that the kth row contains k numbers. Suppose, starting at the top of the triangle, we have to move to adjacent numbers in the row below such that the sum of the selected numbers is maximum.

For example, in the triangle below, the maximum total from top to bottom is 23 given by the path of the underlined numbers, i.e. $3 + 7 + 4 + 9 = 23$.



Note that there are a total of 2^{k-1} paths (where k is the number of rows in the triangle) from the top to the bottom of the triangle. Therefore, it is not feasible to find the maximum sum that can obtained by checking every path when k is large. Devise a dynamic programming solution to find the maximum possible sum and analyze the running time of your solution.

[10 points]